ADVANCED OPTICAL MATERIALS

Supporting Information

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Coherent-Resonance Enhancement of Sensing at the Exceptional Points

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1) Two-modes-approximation model

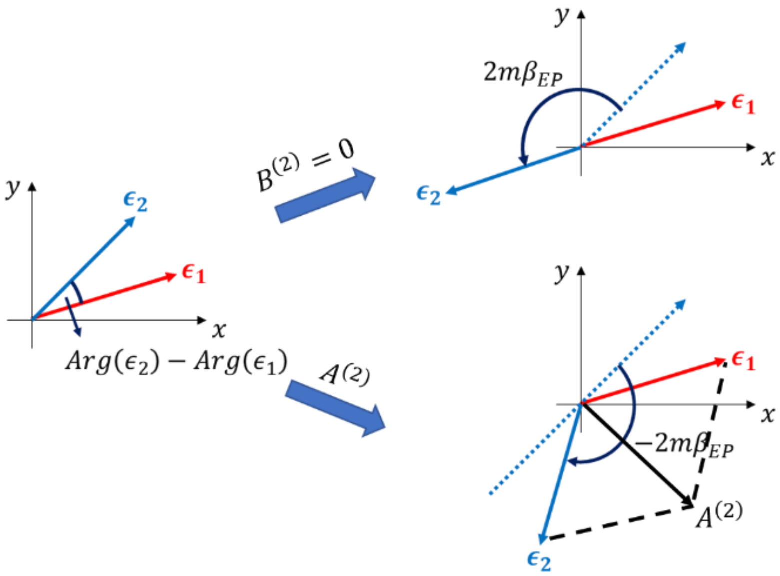


Figure S1. Illustration of conditions for generating EPs

The system consisting of a whispering gallery microcavity with two Rayleigh scatterers (such as particles, nanotips or subwavelength resonators) can be effectively described by 2×2 non-Hermitian Hamiltonian under two-modes approximation as

$$H^{(2)} = \begin{pmatrix} \omega^{(2)} & A^{(2)} \\ B^{(2)} & \omega^{(2)} \end{pmatrix} \tag{S1}$$

with

$$\begin{aligned} \omega^{(2)} &= \omega_0 + \epsilon_1 + \epsilon_2 \\ A^{(2)} &= \epsilon_1 + \epsilon_2 e^{-i2m\beta_2} \\ B^{(2)} &= \epsilon_1 + \epsilon_2 e^{+i2m\beta_2} \end{aligned} \tag{S2}$$

where ω_0 and m are the resonant frequency and the azimuthal number of unperturbed characteristic mode, respectively, ϵ_1 (ϵ_2) is half of the complex modes splitting perturbed by scatterer1 (2) alone, and β_2 is the relative angular position between two scatterers. As claimed in the main text, this system will operate at an exceptional point when $A^{(2)} \neq 0, B^{(2)} = 0$ or $A^{(2)} = 0, B^{(2)} \neq 0$. As illustrated in Figure S1, the equivalent condition for implementing an EP with $A^{(2)} \neq 0$ and $B^{(2)} = 0$ is

$$|\epsilon_1| = |\epsilon_2|,$$
$$Arg(\epsilon_2) - Arg(\epsilon_1) + 2m\beta_{EP} = \pi \tag{S3}$$

And the amplitude of corresponding intrinsic backscattering is given by

$$|A^{(2)}| = 2|\epsilon_1|\sin\big(Arg(\epsilon_2) - Arg(\epsilon_1)\big) \tag{S4}$$

Particularly, we need to point out that the two scatterers are just slightly different in structure and size, meaning that the difference between $Arg(\epsilon_2)$ and $Arg(\epsilon_1)$ is small and can be regarded as a constant. Thus, the intrinsic backscattering $|A^{(2)}|$ is approximately proportional to $|\epsilon_1|$.

2) Derivation of enhancement factor

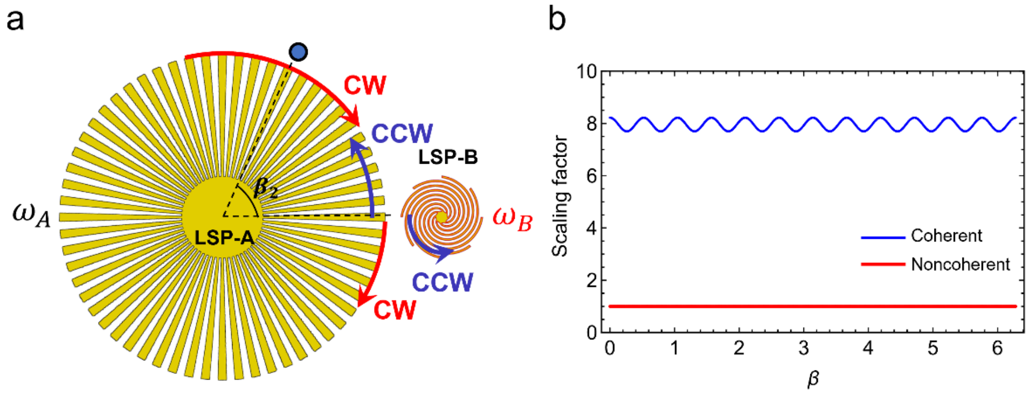


Figure S2. a, Illustration coupled LSPR configuration. b. Numerically calculated ratio factor as a function of angle β for coherent and noncoherent EP-based sensor, respectively. The parameters are set as: $\omega_A = 10$, $\omega_B = \eta * \omega_A$, $g_A = 0.01 + 0.01i$, $g_B = 0.2 + 0.2i$, $m = 6$, $\epsilon_0 = 0.001 + 0.001i$, $\kappa = 0.1$.

We start with the most general case as depicted in **Figure S2a**: a single spiral-shaped LSPR-B with resonance ω_B couples to an azimuthally symmetric LSPR-A with resonance ω_A . This system can be analytically described by the coupled mode theory as:

$$-\frac{da_{CW}}{dt} = i\omega_A a_{CW} + ig_A a_{CW} + ig_A a_{CCW} + i\kappa b_{CCW} \tag{S5.a}$$

$$-\frac{da_{CCW}}{dt} = i\omega_A a_{CCW} + ig_A a_{CCW} + ig_A a_{CW} + i\kappa b_{CW} \tag{S5.b}$$

$$-\frac{db_{CW}}{dt} = i\omega_B b_{CW} + ig_B b_{CW} + ig_B b_{CCW} + i\kappa a_{CCW} \tag{S5.c}$$

$$-\frac{db_{CCW}}{dt} = i\omega_B b_{CCW} + ig_B b_{CCW} + ig_B b_{CW} + i\kappa a_{CW} \qquad (S5.d)$$

In this set of equations, κ is the coupling rate between two LSP resonators. In the LSPR-A (LSPR-B), g_A (g_B) is the coupling rate from CCW to CW and CW to CCW waves, which exist due to the presence of the LSPR-B (LSPR-A) which works as a scatterer. Since LSPR-B is much smaller than LSPR-A, we can infer that g_A is considerably smaller than $g_B (g_A < g_B)$. The corresponding Hamiltonian matrix of Eq. (S5) reads:

$$H = \begin{pmatrix} \omega_A + g_A & g_A & 0 & \kappa \\ g_A & \omega_A + g_A & \kappa & 0 \\ 0 & \kappa & \omega_B + g_B & g_B \\ \kappa & 0 & g_B & \omega_B + g_B \end{pmatrix} \qquad (S6)$$

After some mathematical manipulation, we can get the following eigenvalues:

$$E_1 = \frac{1}{2}\left(\omega_A + \omega_B - \sqrt{(\omega_A - \omega_B)^2 + 4\kappa^2}\right) \qquad (S7.a)$$

$$E_2 = g_A + g_B + \frac{1}{2}\left(\omega_A + \omega_B - \sqrt{(2g_A - 2g_B + \omega_A - \omega_B)^2 + 4\kappa^2}\right) \qquad (S7.b)$$

$$E_3 = \frac{1}{2}\left(\omega_A + \omega_B + \sqrt{(\omega_A - \omega_B)^2 + 4\kappa^2}\right) \qquad (S7.c)$$

$$E_4 = g_A + g_B + \frac{1}{2}\left(\omega_A + \omega_B + \sqrt{(2g_A - 2g_B + \omega_A - \omega_B)^2 + 4\kappa^2}\right) \qquad (S7.d)$$

Note that only two out of these four eigenvalues are possible to degenerate as exceptional points through parameter sweep, either E_1 and E_2 or E_3 and E_4. Here we consider the former case. In this case, the exceptional point is formed by the coalescence of CW and CCW waves within the LSPR-A. Therefore, the effective complex frequency splitting of LSPR-A induced by the presence of LSPR-B is given by:

$$\begin{aligned} 2\epsilon_1(\eta) &= E_2 - E_1 \\ &= g_A + g_B - \frac{1}{2}\sqrt{\left(2g_A - 2g_B - (\eta * \omega_A - \omega_A)\right)^2 + 4\kappa^2} + \frac{1}{2}\sqrt{(\eta * \omega_A - \omega_A)^2 + 4\kappa^2} \end{aligned} \qquad (S8)$$

For fully noncoherent case ($\eta = \infty$), Eq. (R4) reduces to $2\epsilon_1(\infty) = 2g_A$, to in agreement with previous work [13,14]. While for fully coherent case ($\eta = 1$), $2\epsilon_1(1) = g_A + g_B - \sqrt{(g_A - g_B)^2 + \kappa^2} + \kappa$, which is always larger than $2\epsilon_1(\infty)$.

When another target scatterer enters into the evanescent field region to disturb the LSPR-A, the systematic Hamiltonian become:

$$H' = \begin{pmatrix} \omega_A + g_A + \epsilon_0 & g_A + \epsilon_0 e^{-i2m\beta_0} & 0 & \kappa \\ g_A + \epsilon_0 e^{i2m\beta_0} & \omega_A + g_A + \epsilon_0 & \kappa & 0 \\ 0 & \kappa & \omega_B + g_B & g_B \\ \kappa & 0 & g_B & \omega_B + g_B \end{pmatrix} \qquad (S9)$$

To the first-order approximation, the effective complex frequency splitting ϵ_1' of LSPR-A is given by the first two eigenvalues of Eq. (S9), and the approximated Hamiltonian describing these two modes can be written as:

$$H_1' = \begin{pmatrix} \omega_A + {\epsilon_1}'(\eta) & {\epsilon_1}'(\eta) \\ {\epsilon_1}'(\eta) & \omega_A + {\epsilon_1}'(\eta) \end{pmatrix} \tag{S10}$$

This matrix can be broken down into two parts:

$$H_1' = \begin{pmatrix} \omega_A + \epsilon_1(\eta) & \epsilon_1(\eta) \\ \epsilon_1(\eta) & \omega_A + \epsilon_1(\eta) \end{pmatrix} + \alpha \begin{pmatrix} \epsilon_0 & \epsilon_0 e^{-i2m\beta_0} \\ \epsilon_0 e^{+i2m\beta_0} & \epsilon_0 \end{pmatrix} \tag{S11}$$

where α is a scaling factor.

Through numerical calculations, we find that for $\eta = \infty$, the factor $\alpha = 1$, while for $\eta = 1$, the factor $\alpha > 1$, as shown in **Figure S2b.** These results can be interpreted as following:

(1) When the scatterer is fully noncoherent with the WGMs, the coupling rate κ is extremely weak compared to the giant frequency mismatch, so the 4-by-4 matrix equation (S5) can be decoupled into two 2-by-2 matrixes, $\begin{pmatrix} \omega_A + g_A + \epsilon_0 & g_A + \epsilon_0 e^{-i2m\beta_0} \\ g_A + \epsilon_0 e^{i2m\beta_0} & \omega_A + g_A + \epsilon_0 \end{pmatrix}$ and $\begin{pmatrix} \omega_B + g_B & g_B \\ g_B & \omega_B + g_B \end{pmatrix}$, the former is what we use to achieve EP. It can be rewritten as $\begin{pmatrix} \omega_A + g_A & g_A \\ g_A & \omega_A + g_A \end{pmatrix} + \begin{pmatrix} \epsilon_0 & \epsilon_0 e^{-i2m\beta_0} \\ \epsilon_0 e^{+i2m\beta_0} & \epsilon_0 \end{pmatrix}$, indicating that the factor α is naturally equal to 1.

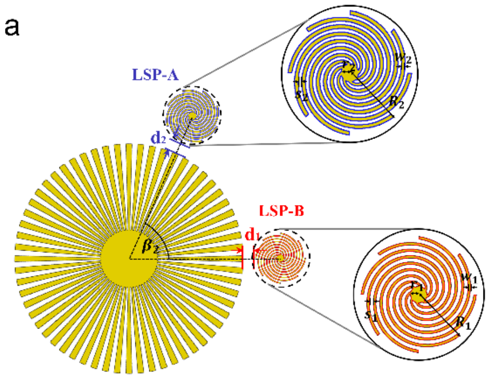
(2) When the scatterer is fully coherent with the WGMs, the coupling rate κ is significantly enhanced compared to the zero-frequency mismatch, so the 4-by-4 matrix equation (S5) cannot be decoupled into 2-by-2 matrix, and the strong coupling enhances the factor α to a greater extent.

As for our proposed structures in the main text with another spiral-shaped LSPR-C, in principle, we can solve the 6-by-6 matrix and construct the effective two-modes approximated Hamiltonian H_2' in the same manner. The 2-by-2 matrix H_2' also can be rewritten as:

$$H_2' = \begin{pmatrix} \omega_A + \epsilon_1(\eta) + \epsilon_2(\eta) & \epsilon_1(\eta) + \epsilon_2(\eta) e^{-i2m\beta_2} \\ \epsilon_1(\eta) + \epsilon_2(\eta) e^{+i2m\beta_2} & \omega_A + \epsilon_1(\eta) + \epsilon_2(\eta) \end{pmatrix} + \alpha \begin{pmatrix} \epsilon_0 & \epsilon_0 e^{-i2m\beta_0} \\ \epsilon_0 e^{+i2m\beta_0} & \epsilon_0 \end{pmatrix} \tag{S12}$$

where α is enhancement factor, $\alpha = 1$ for $\eta = \infty$ and $\alpha > 1$ for $\eta = 1$.

3) Detailed design of the simulated three EP-based sensors



b

η	LSP-A R_1	r_1	w_1	s_1	d_1	LSP-B R_2	r_2	w_2	s_2	d_2
∞	2.6	-	-	-	0.5	2.8	-	-	-	0.51
1.05	2.6	0.3828	0.18	0.1225	0.5	2.8	0.5916	0.2	0.1514	0.6
1	2.6	0.3817	0.14	0.1557	0.5	2.8	0.4924	0.2	0.1333	0.51

Parameters unit: mm

Figure S3. a, Schematic of proposed EP-based sensor. b, The optimized parameters of spiral-shaped LSPs of three EP-based sensors with degrees of resonance coherence, $\eta = \infty$, 1.05, 1, respectively.

Figure S3a displays the detailed 2D model of the EP-based sensor investigated in the main text. The spiral-shaped resonator LSP-A (LSP-B) is composed of an inner disk of radius $r_1(r_2)$ surrounded by eight spiral shaped gooves with radius $R_1(R_2)$, with the groove spacing $s_1(s_2)$ and groove width $w_1(w_2)$. We focus on their dipole modes and adjust the parameters precisely to realize EP-based sensors with different degrees of resonance coherence. **Figure S3b** shows the parameters of spiral-shaped LSPs of three EP-based sensors with different degrees of resonance coherence, $\eta = \infty$, 1.05, 1, respectively, as demonstrated in **Figure 2**.

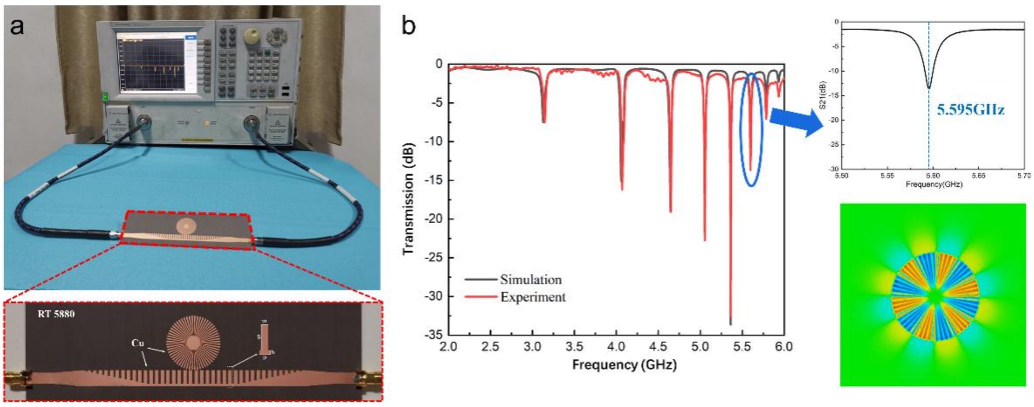


Figure S4. a, Photograph of the sample and the experimental setup for the transmission spectrum measurement. b, Measured and simulated transmission spectrum. The inset shows the transmission spectrum around the sixth-order resonance mode and the corresponding near-field distribution ($|E_z|^2$).

4) Experimental setup

Our designed sample structure includes a DP-based/EP-based sensor and an SSPP waveguide, fabricated with 18μm thick copper printed on a 1.575mm thick RT5880 (permittivity $\epsilon = 2.2 \pm 0.02$, loss tangent $\Delta\delta = 0.0009$) dielectric substrate. Both ends of this waveguide are connected to a vector network analyzer via SMA connectors to excite multi-modes as demonstrated by several dips in the transmission spectrum **Figure S4b**. Here, we select the sixth-order mode at 5.595GHz as the characteristic mode for further adjustment to realize an exceptional point.

5) Spectral identifications of exceptional points of EP-based sensors

In our simulation designs, we identify the exceptional points of practical fully non-coherent and fully coherent EP-based sensors by adjusting the relative phase angle β_2 between the particles or spiral-shaped LSP resonators. As shown by transmission spectrums in **Figure S5**, when β_2 increases continuously, the two split modes merging gradually and coalesce at a certain angle β_{EP} (marked in red), which indicates the realization of exceptional points. Then the modes bifurcate again when β_2 is increased further.

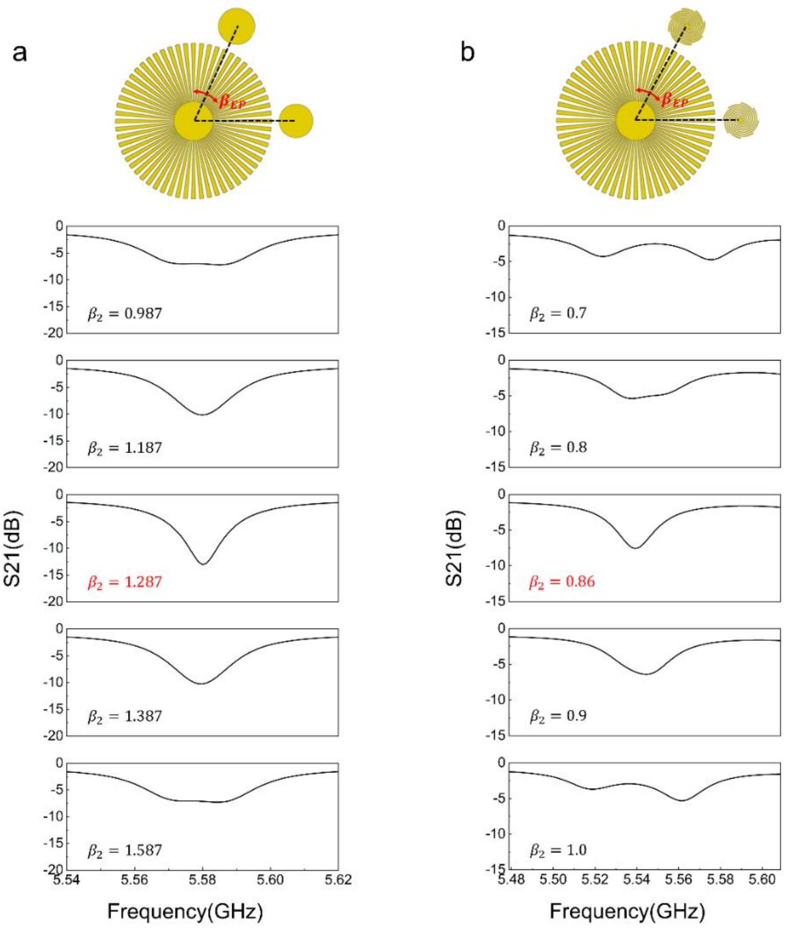


Figure S5. Simulated transmission spectra as the relative phase angle β_2 between the particles or spiral-shaped LSPs is varied for fully non-coherent (a) and fully coherent (b) EP-based sensors. The mode coalescence can be clearly observed in the middle panels (marked in red).

6) Resolvability analyses of EP-based sensors

We define the resolvability of sensing as: $R_s = \frac{2\Delta f}{\gamma_1+\gamma_2}$, where Δf is the frequency split of resonance dips, γ_1 and γ_2 designate the linewidths of the resonances, as illustrated in **Figure S6a**. we plot the resolvability as a function of perturbation strength for both noncoherent and coherent EP-based sensors, as shown in **Figure S6c**. Apparently, the resolvability increases when the perturbation strength increases. It is worth noting that for non-coherent EP sensors, the resolvability is smaller than 1. This result agrees with the previous study which shows that, in the close proximity of EP_2, the enhanced noise saturates SNR and hence the resolvability. Nonetheless, the coherent resonance provides a possible route to increase the resolvability and all these data represent good sensing spectrum resolvability.

Specifically, we depict the experimental spectra of sensing for coherent and noncoherent EP-based sensors at the weakest perturbation strength. They correspond to the cases with the lowest resolvability in our experiment. One can observe clear splitting in the resonance dips.

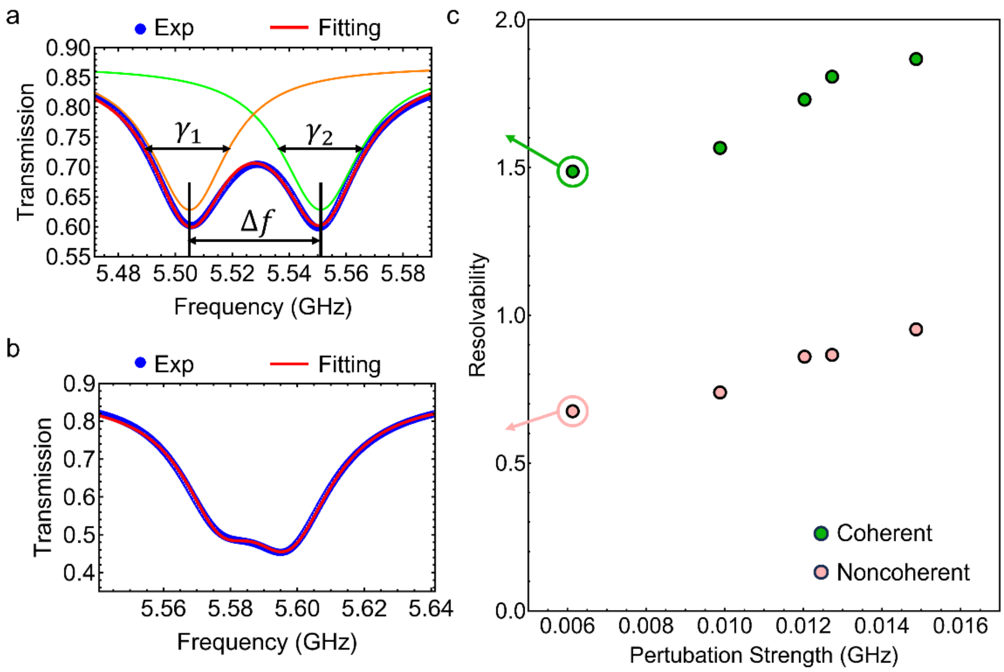


Figure S6. a, Experimental spectrum of sensing for coherent EP-based sensors at the weakest perturbation strength in c. b, Experimental spectrum of sensing for noncoherent EP-based sensors at the weakest perturbation strength in c. c, Sensing resolvability as a function of perturbation strength for coherent (green dots) and noncoherent (pink dots) EP-based sensors.